Expert Systems With Applications 302 (2026) 130478



Contents lists available at ScienceDirect

Expert Systems With Applications

journal homepage: www.elsevier.com/locate/eswa





InvaderDefender: Multimodal recognition of invasive alien species via a large language model with vision-guided targeted RAG

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ARTICLE INFO

Keywords:
Invasive alien species (IAS)
Multimodal fusion
Large language model
Long-tailed distribution
Retrieval augmented generation (RAG)
Biodiversity monitoring

ABSTRACT

Invasive Alien Species (IAS) pose a significant threat to global biodiversity and economies, necessitating accurate and efficient identification for effective management. Traditional methods are often constrained by time, cost, and expertise while existing unimodal deep learning approaches struggle with species phenotypic diversity, environmental complexity, and long-tailed data distributions. This study introduces InvaderDefender, a novel "Visual Recognition-Semantic Enhancement" two-stage multimodal fusion framework to address these challenges. The first stage utilizes an EfficientNetV2-L visual backbone, enhanced with a novel wavelet fusion module for fine-grained feature extraction, and a combined strategy of class-aware data resampling and gradient-aware focal loss to address long-tailed distributions. The second stage inputs the Top-k probability outputs from the visual model, along with user-provided text, into the GLM-4 LLM enhanced by our proposed Vision-Guided Targeted Retrieval Augmented Generation (VGT-RAG). This GLM-4 leverages a specially constructed expert knowledge base for 20 target IAS to perform in-depth inference and output the final identification. To support this research, the first multimodal benchmark dataset specifically for these IAS was constructed. InvaderDefender achieves a Top-1 accuracy of 99.31 %, significantly outperforming mainstream unimodal models. Compared to the visual-only baseline, the full multimodal framework boosts Top-1 and Mean Accuracy by 0.90 and 2.15 percentage points, respectively, ensuring a more balanced performance across all species, particularly rare ones. This work establishes an efficient and robust framework for IAS identification, providing a novel methodology for integrating multimodal learning and knowledge-driven AI in biodiversity monitoring.

1. Introduction

Invasive Alien Species (IAS) are an escalating global threat, undermining biodiversity, disrupting ecosystem functions, and damaging regional agricultural economies. These species are estimated to cause annual economic losses totaling several trillion U.S. dollars, jeopardizing global food security, public health, and progress toward sustainable development goals (Roy et al., 2024). Early and accurate identification of IAS is therefore critical for timely and effective prevention and control measures that can minimize both ecological and economic damage.

Traditional IAS monitoring and identification methods rely heavily on field surveys and morphological taxonomy by expert identifiers, occasionally supplemented with DNA barcoding for molecular verification (Hebert et al., 2003). While morphological methods can be highly accurate under ideal conditions, they depend heavily on expert knowledge and are limited in handling high phenotypic plasticity or ambiguous morphological traits (Pyšek et al., 2013). DNA barcoding helps address some of these issues but remains constrained by high laboratory costs and incomplete reference databases (Demertzis & Iliadis, 2017). Overall, these approaches face challenges of high labor intensity, which primarily stems from the dependency on specialized taxonomic expertise for morphological analysis, low scalability, and inefficiency in high-frequency or large-scale monitoring-highlighting the urgent need for automated and intelligent identification solutions.

Recent advances in computer vision, particularly through Deep Learning (DL) and Convolutional Neural Networks (CNNs), have transformed species recognition (LeCun et al., 2015; Shoaib et al., 2023). DL models automatically extract hierarchical visual features from raw images, significantly outperforming traditional machine learning

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https://doi.org/10.1016/j.eswa.2025.130478
Received 17 June 2025; Received in revised form 21 October 2025; Accepted 16 November 2025
Available online 27 November 2025
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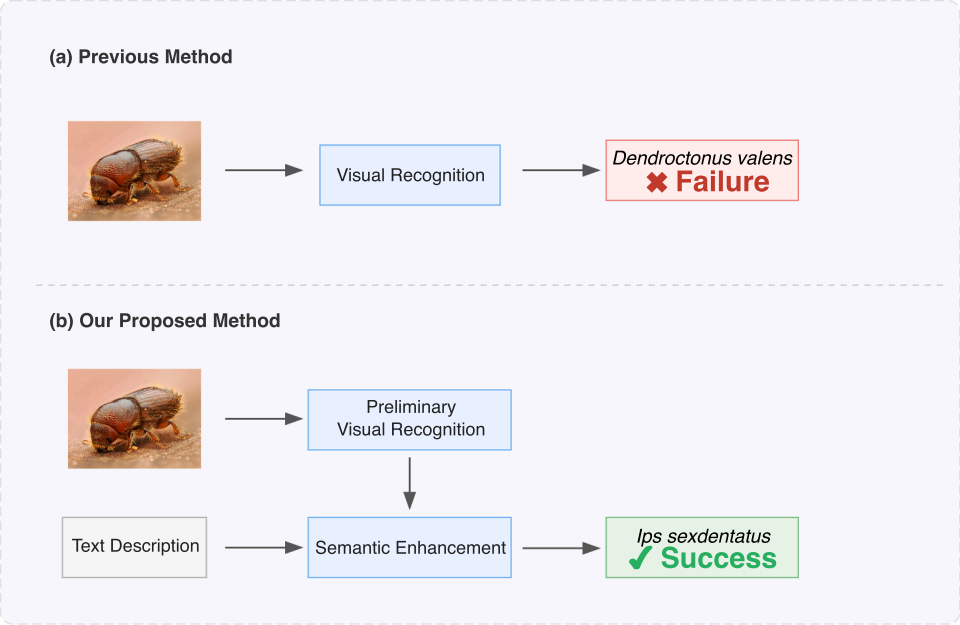


Fig. 1. Conceptual diagram of the proposed multimodal framework compared to a conventional unimodal method. (a) The conventional method, relying solely on visual recognition, may fail to distinguish between morphologically similar species, leading to an incorrect identification. (b) Our proposed method integrates preliminary visual recognition with semantic enhancement from a text description, allowing it to resolve visual ambiguity and achieve a correct identification.

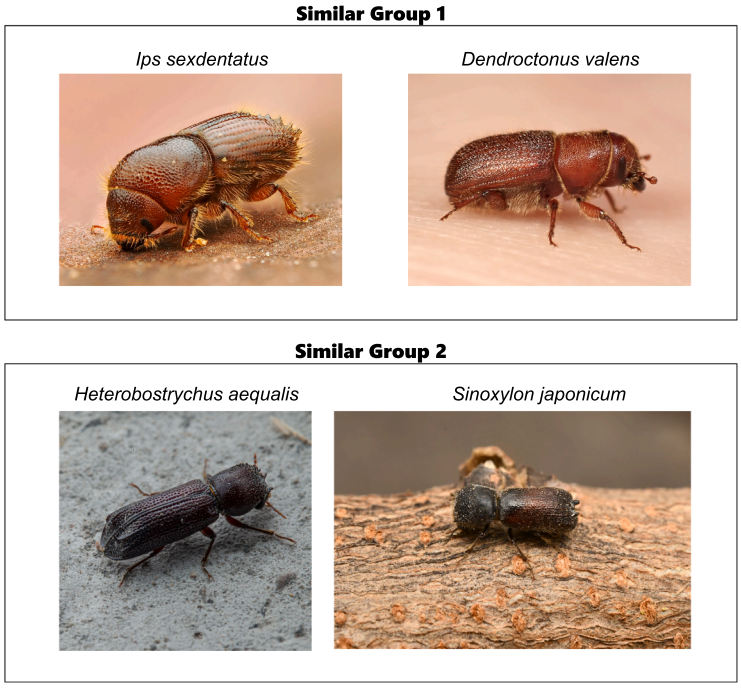


Fig. 2. Examples of high inter-species visual similarity that pose challenges for automated identification. The figure displays two pairs of morphologically similar beetles:

approaches that rely on manual feature engineering (Gao et al., 2024; Kotsiantis et al., 2007). These models have demonstrated strong potential in identifying invasive plants, pests, and aquatic species (Jensen et al., 2020; Shoaib et al., 2023).

However, deploying DL-based IAS identification systems in real-world scenarios still faces several critical obstacles:

1. **Limitations of visual-only recognition in complex environments:** Relying solely on image data is often insufficient in the face of high intra-species phenotypic plasticity, inter-species visual similarity (e.g., between *Dendroctonus valens* and *Ips sexdentatus*, as shown in Fig. 2), and complex environmental backgrounds. Such unimodal systems lack the contextual and domain-specific reasoning that human experts naturally apply (Wei et al., 2021).
2. **Challenges in fine-grained classification of morphologically similar species:** Many IAS are notoriously difficult to distinguish from native or non-target alien species due to exceedingly subtle morphological differences-such as minor variations in appendage structure, color patterning, or surface texture. Conventional visual models, typically optimized for general object recognition, often

lack the architectural sensitivity or training mechanisms to prioritize these diagnostically critical cues. As a result, they frequently misclassify morphologically similar species, undermining precise detection and effective management efforts.

3. **Dataset scarcity and imbalance in IAS datasets:** Training robust IAS models faces a dual challenge. First, there is a significant scarcity of public, high-quality benchmark datasets for this task (Sun et al., 2017). Specifically, a comprehensive multimodal dataset that pairs species images with structured, annotated text is notably absent, hindering the development of advanced, context-aware models. Second, the data that can be compiled, including the benchmark dataset constructed for this study, exhibits severe imbalance, with an overrepresentation of common species and minimal samples for many others, such as *Rhabdoscelus lineaticollis* or *Brontispa longissima*. This long-tail imbalance leads to biased predictions and poor generalization on these underrepresented classes (Lin et al., 2017).

To address these challenges, we propose **InvaderDefender**, a novel two-stage multimodal identification framework that integrates visual recognition with expert-driven semantic reasoning, mimicking the comprehensive decision-making process of human experts, as conceptually illustrated in Fig. 1. This study aims to automate the identification process, thereby significantly reducing the dependency on taxonomic experts and shortening the identification time from days to seconds. The user's manual operation is limited to capturing clear photographs and providing optional auxiliary descriptions. This approach greatly broadens the participation in IAS monitoring, enabling the public and grassroots forestry workers to become an effective monitoring force, thereby achieving large-scale, high-frequency monitoring and solving the bottleneck of the traditional expert-identification model.

Our contributions are as follows:

1. Proposing a novel "Visual Recognition-Semantic Enhancement" framework that mimics expert decision-making: We innovatively designed the InvaderDefender, a two-stage framework that emulates the cognitive process of a human expert. The first stage employs a wavelet-enhanced visual model to perform precise, fine-grained feature extraction, akin to an expert's initial visual inspection. The second stage then pioneers the use of the visual model's outputs to guide a Retrieval Augmented Generation (RAG) system, which consults a specialized knowledge base for in-depth reasoning-mirroring an expert's reliance on accumulated knowledge to confirm identification. This synergistic architecture overcomes the limitations of unimodal systems, significantly enhancing both accuracy and robustness.
2. **Constructing the first multimodal benchmark dataset and expert knowledge base for key IAS:** To facilitate this research and fill a critical resource gap in the field, we developed the first high-quality multimodal benchmark dataset for 20 key invasive alien species. This dataset pairs annotated images with structured, expert-authored textual descriptions. Crucially, we also compiled a comprehensive expert knowledge base, which serves as the essential foundation for our framework's semantic enhancement stage, providing the indispensable domain-specific knowledge required for high-level reasoning.
3. **Developing a comprehensive strategy to address the critical challenge of long-tailed data distribution:** Recognizing that real-world IAS data is inherently imbalanced, we implemented and validated a combined strategy of class-aware data resampling and gradient-aware Focal Loss. This approach directly tackles the long-tailed distribution problem, a major bottleneck for training robust models. Our experiments show it significantly improves identification performance for rare and under-sampled species, ensuring more equitable and reliable results in practical application scenarios.
4. **Pioneering a vision-guided targeted RAG (VGT-RAG) mechanism for biological identification:** We introduce a groundbreaking VGT-RAG mechanism that fundamentally improves upon conventional RAG systems. Instead of relying on potentially ambiguous user text, our method uses the high-confidence Top-k predictions from the visual model as a precise "navigational signal" for targeted information retrieval from the expert knowledge base. This ensures the Large Language Model receives highly relevant and accurate context, drastically enhancing its reasoning capabilities, especially when distinguishing between morphologically similar species. This offers a new, more effective paradigm for multimodal fusion in fine-grained classification tasks.

2. Related work

The identification of Invasive Alien Species (IAS) has evolved from expert-driven morphological methods to advanced computational techniques. This section reviews key developments in IAS identification, highlights persistent limitations, and motivates the design of our multimodal framework, InvaderDefender.

2.1. Traditional and early computational approaches

Historically, IAS identification relied on morphological analysis by taxonomic experts and, later, DNA barcoding for molecular confirmation (Hebert et al., 2003; Pyšek et al., 2013). Morphological methods depend on expert knowledge to compare external traits (e.g., leaf shape in plants, body morphology in insects), but they are constrained by phenotypic plasticity, cryptic species, and a global shortage of taxonomists, limiting rapid, large-scale identification (Hünicken et al., 2024; Pyšek et al., 2013). DNA barcoding-typically using mitochondrial COI for animals-improves precision and can identify cryptic or immature specimens (Hebert et al., 2003; Hünicken et al., 2024). However, its high laboratory costs, incomplete reference databases, and processing time hinder real-time, on-site monitoring (Demertzis & Iliadis, 2017; Pyšek et al., 2013).

To alleviate these constraints, early machine learning (ML) methods were adopted to automate aspects of IAS identification. Traditional algorithms-such as Support Vector Machines (SVM), Random Forests (RF), and Artificial Neural Networks (ANN)-were applied to features extracted from images, environmental DNA (eDNA), or remote sensing data (Kotsiantis et al., 2007). For example, Demertzis and Iliadis (2017) proposed a semi-supervised Extreme Learning Machine combined with a spiking neuron model to identify invasive pufferfish via eDNA (Demertzis & Iliadis, 2017). Xiao et al. (2018) compared SVM, RF, and ANN for predicting aquatic invasive species eradication success, finding RF most effective at identifying key ecological factors (Xiao et al., 2018). Remote sensing combined with ML also showed promise: Crisigiovanni et al. (2021) used multispectral WorldView-2 imagery and RF to distinguish endangered *Araucaria angustifolia* from invasive *Hovenia dulcis*, achieving an F_1-score of 95 % (Crisigiovanni et al., 2021). Similarly, Jensen et al. (2020) employed RF and SVM on Sentinel-2 and AVIRIS hyperspectral data to detect invasive Kudzu in the southeastern United States (Jensen et al., 2020). Despite these successes, traditional ML models depend heavily on manual feature engineering-requiring expert knowledge to design discriminative features-and often struggle to capture subtle visual cues necessary for distinguishing morphologically similar IAS (Gao et al., 2024). Consequently, issues such as limited generalization, sensitivity to complex backgrounds, and inability to handle imbalanced data remained.

2.2. Deep learning for IAS identification: advances and persistent challenges

Deep Learning (DL), particularly Convolutional Neural Networks (CNNs), marked a significant advancement by enabling automatic hierarchical feature extraction from raw imagery (LeCun et al., 2015). Transitioning from traditional ML, DL models eliminate much of the manual feature engineering burden and boast superior performance in visual tasks (Gao et al., 2024). For instance, Cabezas et al. (2020) employed ResNet and ResNeXt architectures on high-resolution UAV imagery to

detect highly imbalanced invasive blueberry species in wetlands, using weighted loss functions and data augmentation to achieve robust detection (Cabezas et al., 2020). Lake et al. (2022) combined WorldView-2 and PlanetScope imagery with CNN and LSTM networks to detect leafy spurge (*Euphorbia virgata*), demonstrating improved accuracy by incorporating phenological information (Lake et al., 2022). In another example, Martinez et al. (2021) compared YOLOv4 and SSD-Mobilenet-v2 for real-time detection of invasive lionfish using Autonomous Underwater Vehicles, confirming DL's feasibility in complex marine environments (Martínez-González et al., 2021).

Despite these advances, DL-based IAS identification still faces three persistent challenges:

1. **Fine-grained visual classification (FGVC):** Many IAS closely resemble native or other alien species, differing only in minute morphological details. While some DL architectures introduce attention mechanisms or part-localization strategies (Wei et al., 2021; Xie et al., 2023), relying solely on visual features remains insufficient for capturing these subtle cues. Our InvaderDefender framework addresses this by integrating a wavelet decomposition module to enhance multi-scale, textural feature extraction (Section 4.2.3).
2. **Long-tailed data distributions:** Real-world IAS datasets exhibit severe class imbalance-common species have abundant samples, whereas rare or recently emerged IAS have very few (Sun et al., 2017). Models trained on such data tend to be biased toward majority classes. Techniques like Focal Loss (Lin et al., 2017) and class-aware resampling can mitigate imbalance; our work combines both to improve rare species recognition (Section 4.2.2) (Cabezas et al., 2020; Chen et al., 2024).
3. **Limitations of unimodal visual recognition:** Advanced DL models still ignore contextual information-such as habitat, behavior, or geographical distribution-that experts use when identifying IAS. Consequently, DL systems may misclassify species that are visually similar but ecologically distinct. This underscores the need for multimodal approaches that fuse visual and textual or geospatial data. Recent studies have pointed to the promise of combining visual with contextual information (Cardoso et al., 2024), motivating our use of semantic enhancement via expert knowledge.

2.3. Large language models (LLMs) and RAG in IAS identification

Large Language Models (LLMs) have revolutionized natural language processing by demonstrating strong capabilities in contextual understanding and reasoning (GLM et al., 2024). Retrieval Augmented Generation (RAG) further empowers LLMs by enabling dynamic access to external knowledge bases, improving factual grounding and relevance (Gupta et al., 2024; Lewis et al., 2020). In domains such as medical diagnosis, RAG systems have been used to retrieve pertinent clinical guidelines to assist decision-making (Jia et al., 2025).

In IAS identification, integrating LLMs could help incorporate morphological descriptions, ecological traits, and distribution data-information that visual models alone cannot capture. However, existing RAG pipelines typically rely on text-based user queries to retrieve from large, general knowledge repositories. Such general retrieval may yield overly broad or less relevant information, particularly when user descriptions are imprecise or when distinguishing among morphologically similar species is critical. For example, retrieving generic "beetle morphology" content will not effectively differentiate *Dendroctonus valens* from *Ips sexdentatus*. This gap motivates more precise retrieval strategies.

Our work introduces a Vision-Guided Targeted RAG (VGT-RAG) mechanism (Section 4.3), where the visual model's Top-k predictions guide retrieval from a specialized IAS expert knowledge base. By using high-confidence visual cues (e.g., likely genera or species groups) as the retrieval seed, VGT-RAG ensures that the LLM receives highly relevant, domain-specific information-enhancing fine-grained discrimination and compensating for data imbalance. This targeted approach represents a novel application of RAG in biological classification and addresses the limitations of purely text-driven retrieval.

3. Dataset

3.1. Image acquisition and composition

The dataset utilized in this study is meticulously compiled and annotated by the authors. Image data are primarily sourced from established public databases, namely the Global Biodiversity Information Facility (GBIF) and the Centre for Agriculture and Bioscience International (CABI). This collection is supplemented with specimen images photographed under laboratory conditions. The dataset comprises a total of 11,633 images representing 20 key invasive alien species (IAS). Each image is accompanied by corresponding textual annotations. A detailed breakdown of the image count per species is presented in Table 1, and the class distribution is visualized in Fig. 3.

3.2. Multimodal textual annotations

To facilitate multimodal learning and robust evaluation, structured textual information is developed to correspond with each image in the dataset. This information includes systematically curated user descriptions (`user_input`), designed to mimic submissions from a mobile application, alongside standardized species descriptions (`standard_input`) reserved for future research endeavors. Each entry is organized in JSON format and contains the following key fields:

A concrete visualization of this data structure is provided in Fig. 4. The fields are defined as follows:

- `image_id`: A unique identifier linking the text entry to its corresponding image file.
- `species_name`: The scientific (Latin) name and common Chinese name of the species.
- `description`: A structured compilation of detailed species information, including nested subfields for morphology, ecology, distribution, and potential impact. The content for these fields is curated from the *IAS Expert Knowledge Base* (Section 3.3) . This component is primarily intended to support the future development of end-to-end multimodal models capable of in-depth species knowledge comprehension; consequently, it was not utilized in the experiments for the current framework.
- `user_input`: Text curated by trained annotators, based on expert-defined guidelines, to simulate the descriptions a user might provide alongside a photograph in a real-world submission. These guidelines specified creating descriptions with varying levels of detail, using non-technical language, and focusing on different visual cues (e.g., only color, only shape) to ensure diversity. The resulting descriptions therefore include potential ambiguities and incomplete information to reflect real-world scenarios. To maintain high quality and consistency, a subset of these descriptions was reviewed by a domain expert to verify adherence to the guidelines. In the current framework, this field provides the user's textual description for the final LLM-based reasoning stage. During evaluation, it is also used to assess the model's comprehensive identification capabilities. In the experiments conducted in this study, this field served as the **sole source** of user-provided text (x_T) for the Semantic Enhancement Module (Section 4.3).
- `standard_input`: A standardized textual description reserved for future training of native end-to-end multimodal models designed to foster alignment between visual and textual features. Unlike the `user_input`, which simulates subjective, non-expert observations, this field is intended to contain objective, expert-verified descriptions of the image content. As this field is reserved for future work, it was **not utilized** in any of the experiments or modules of the InvaderDefender framework presented in this study.

Table 1
Per-species image counts revealing significant class imbalance. The table lists the exact number of images for each of the 20 species, with counts ranging from 53 for *Rhabdoscelus lineaticollis* to 2688 for *Obolodiplosis robiniae*.

Species Name	Number of Images	Species Name	Number of Images
Aulacaspis yasumatsui	87	*Hyphantria cunea*	516
Leptocybe invasa	253	*Icerya aegyptiaca*	855
Sinoxylon japonicum	68	*Obolodiplosis robiniae*	2688
Acanthoscelides macrophthalmus	77	*Corythucha ciliata*	1064
Hyphantria cunea larva	1002	*Heterobostrychus aequalis*	172
Brontispa longissima	64	*Icerya purchasi*	946
Ips sexdentatus	106	*Rhynchophorus ferrugineus*	923
Dendroctonus valens	636	*Lantana camara*	1028
Hylurgus ligniperda	220	*Phenacoccus solenopsis*	101
Sirex noctilio	774	*Rhabdoscelus lineaticollis*	53

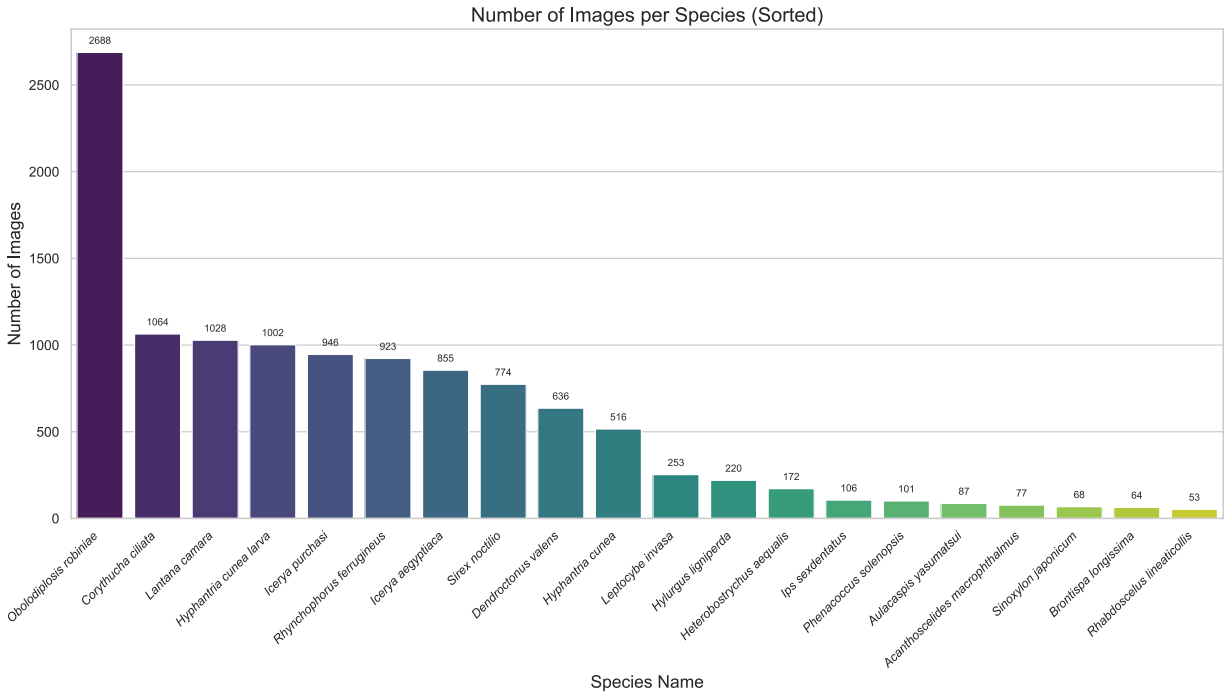


Fig. 3. Visualization of the dataset's long-tailed distribution. This histogram plots the number of images per species (detailed in Table 1), visually confirming the severe data imbalance where a few "head" classes have abundant samples while the majority of "tail" classes are underrepresented.

This structured textual data furnishes the model with rich semantic information, a critical factor for achieving effective multimodal fusion.

3.3. IAS expert knowledge base

To underpin the semantic enhancement stage of the proposed framework, an expert knowledge base for the 20 target IAS was meticulously constructed following a rigorous, three-part protocol. First, for **source selection**, the knowledge base was compiled from two primary types of authoritative sources: (1) official standards and technical documents issued by governmental bodies like the National Forestry and Grassland Administration of China, and (2) peer-reviewed academic literature. A systematic search was conducted on major scientific databases (e.g., Web of Science, Google Scholar, and CNKI) using keywords that included the species' **scientific names** (e.g.,*Dendroctonus valens*), **common names**, and specific terms such as "morphology", "ecology", and "control". The selection criteria prioritized comprehensive review articles and foundational taxonomic studies. Second, to ensure inter-species consistency, a standardized **annotation schema (ontology structure)** was established. For each species, information was organized into a consistent template covering key fields such as: `taxonomic details`, `morphology` (including characteristics of different life stages like egg, larva, pupa, and adult), `ecology` (covering life cycle, host plants, and habitat), `geographical distribution`, `ecological impact`, and established `control measures` (including chemical, physical, and biological methods). Finally, to ensure **accuracy and completeness**, information extracted from multiple sources was cross-validated. Descriptive traits were synthesized from a consensus of findings, and any discrepancies were resolved by deferring to the official governmental standards. This systematic protocol ensures that the resulting knowledge base is reliable, structured, and directly tailored to the task of fine-grained species identification.

4. Methodology

The multi-modal fusion identification framework for Invasive Alien Species (IAS) proposed in this study, named InvaderDefender, adopts a two-stage process: visual recognition followed by semantic enhancement using a Retrieval Augmented Generation (RAG) based Large Language Model (LLM). The overall architecture is illustrated in Fig. 5.

```
{
"image_id": "example_00001",
"species_name": "Sinoxylon japonicum",
"description": {
    "morphology": "Physical characteristics and structural features
         of the organism's different life stages (e.g., adults,
        eggs, larvae, pupae), such as size, color, and anatomical
        structures.",
    "ecology": "Biological and ecological behaviors of the organism
        , including life cycle, habitat preferences, seasonal
        activities, and interactions with the environment.",
    "distribution": "Geographical spread and occurrence patterns of
         the organism, including regions where it has been observed
         or reported.",
    "impact": "Effects of the organism on its environment, host
        species, or human activities, such as ecological damage,
        economic losses, or health impacts."
},
"user_input": "Subjective observations or descriptions provided by
    users, typically including physical traits, collection context,
     discovery locations, and preliminary identifications.",

"standard_input": "standard_input is a text field for detailed,
    standardized descriptions of an image's content, capturing
    visual elements and contextual details to enable semantic
    alignment. It typically includes observations about the subject
    's appearance (e.g., color, shape, texture), actions,
    surroundings, and contextual clues (e.g., habitat,
    environmental conditions), formatted to facilitate machine
    understanding and cross-modality integration."
}
```

Listing 1. Data structure for the multimodal textual annotations. Each JSON entry links an image to its corresponding `species_name`, expert descriptions (e.g., `morphology`, `ecology`), and a simulated `user_input`, providing the rich semantic context required for the InvaderDefender framework.

4.1. Overall framework of InvaderDefender

As illustrated in Fig. 5, our proposed InvaderDefender framework is architected as a two-stage process that emulates an expert's identification workflow: an initial, fine-grained visual assessment followed by a knowledge-driven semantic refinement. This synergistic structure is designed to deliver highly accurate and robust species identification.

The process unfolds as follows:

1. **Stage 1: visual recognition.** The workflow begins with an input image x_I, which is processed by the Visual Recognition Module (M_E). This module is built upon an EfficientNetV2-L backbone and is augmented with two critical enhancements: a **Wavelet Enhance** component to capture subtle, multi-scale textural features, and a **Long-tail Strategy** to effectively address class imbalance within the dataset. The output of this stage is a vector of Top-k class probabilities, P_S. In practical terms, as shown in Eq. 1, this means the Visual Recognition Module (M_E) processes the input image (x_I) to generate an initial visual hypothesis (P_S).

$$P_S = M_E(x_I) \tag{1}$$

2. **Stage 2: semantic enhancement.** This stage refines the initial visual prediction using expert knowledge. The Top-k probabilities P_S from Stage 1, along with any supplementary user-provided text x_T, are fed into the **VGT-RAG Module (M_{RAG})**. The core of this module is its **Vision-Guided Targeted Retrieval** mechanism. Instead of performing a broad search based on user text, it uses the high-probability candidate species from P_S to conduct a precise, targeted query into the **IAS Expert Knowledge Base (K)**. This ensures the retrieved information is highly relevant. The module then integrates these visual probabilities, user text, and retrieved knowledge. As described in Eq. 2, this process means the VGT-RAG Module (M_{RAG}) combines the visual hypothesis (P_S), user text (x_T), and expert knowledge (K) to create the final, enriched context (C_{final}) for the LLM.

$$C_{final} = M_{RAG}(P_S, x_T, K) \tag{2}$$



User description

I found a small black beetle, about 1 cm long, inside a tree trunk in Hebei, China.

Fragment of expert knowledge

The adult beetle is 10–12 mm long, cylindrical, and reddish-brown. Its antennae are light brown with the last three segments comb-like. The pronotum extends forward like a cap, covering the head, and has many granular and serrated projections on the front half. The thorax and abdomen are covered with fine grayish-yellow hairs. The abdomen has five segments, each side with four well-developed reddish-brown serrated spines slightly curved backward. The elytra are rough with brown granules, sloping down at the rear, which is dark brown and very coarse, with a pair of spine-like projections near the rear center.

Description of Modality Alignment

This image shows a small beetle burrowing into wood. The beetle has dark brown elytra with a textured surface and is covered in wood dust. It is inside damaged wood filled with tunnels and powder, indicating insect activity. The background is blurred, suggesting an outdoor setting, while the focus is sharp on the beetle and wood.

Fig. 4. An example of a multimodal data entry for *Sinoxylon japonicum*, showing the correspondence between the species image and key textual annotations extracted from the dataset.

Finally, this comprehensive context C_{final} is passed to the **GLM-4 Large Language Model (M_L)** for final inference. In practice, as shown in Eq. 3, the LLM (M_L) analyzes all the provided evidence within C_{final} to make a definitive judgment and output the final species identification, $\hat{y}$.

$$\hat{y} = M_L(C_{final}) \tag{3}$$

The complete pipeline can be conceptually represented as a sequential flow where the visual model's output actively guides the knowledge retrieval and semantic reasoning process:

$$x_I \xrightarrow{M_E} P_S \xrightarrow[+x_T]{M_{RAG}(\cdot,\cdot,K)} C_{final} \xrightarrow{M_L} \hat{y} \tag{4}$$

This two-stage approach allows InvaderDefender to first leverage the power of deep learning for nuanced visual feature extraction and then employ the reasoning capabilities of an LLM, enriched with expert knowledge, to resolve ambiguity and confirm the identification.

4.2. Visual recognition module (M_E)

The Visual Recognition Module (M_E) is responsible for extracting discriminative visual features from input IAS images and generating an initial probability distribution across target species. As depicted in Fig. 6, it is architecturally composed of a primary EfficientNetV2-L backbone augmented by a parallel wavelet processing branch, with features from both paths merged via a specialized Attention Fusion module.

4.2.1. Backbone network selection

To establish a robust foundation for visual feature extraction, a systematic evaluation of ten mainstream deep learning models was conducted on our IAS dataset. Among these, EfficientNetV2-L demonstrated superior performance, achieving a Top-1 Accuracy of 98.41 %, a Macro F1 Score of 94.96 %, and a Macro Precision of 95.19 % in these prelim-

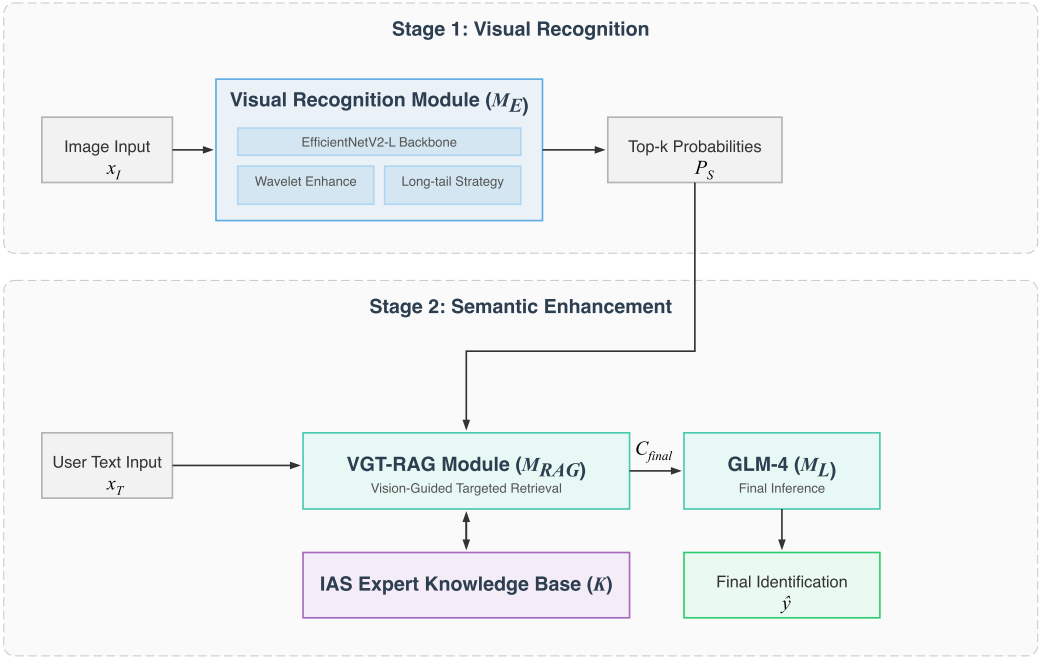


Fig. 5. The overall two-stage architecture of the InvaderDefender framework. It integrates a Visual Recognition module with a Semantic Enhancement module powered by a Vision-Guided Targeted RAG system and a final inference LLM.

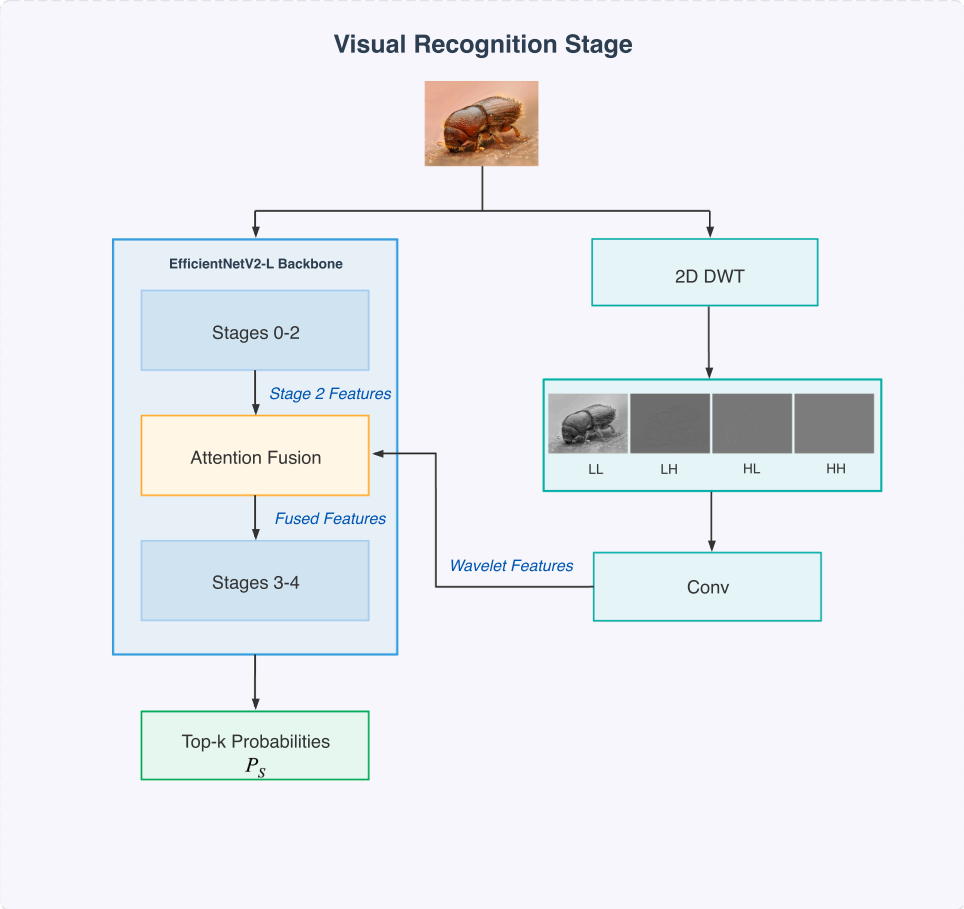


Fig. 6. The architecture of the Visual Recognition Stage (M_E). The model employs a parallel architecture to fuse standard deep features from the EfficientNetV2-L backbone with frequency-domain features derived from a 2D Discrete Wavelet Transform (DWT). This fusion of spatial and frequency information is performed using a dedicated attention module to enhance fine-grained recognition.

inary assessments (detailed comparative results of all evaluated backbones are presented in the Experiments section, Table 2). Consequently, EfficientNetV2-L was selected as the visual backbone network for M_E. The feature extraction and classification process within M_E can be abstractly represented as:

$$P_S = \text{Softmax}(W \cdot \phi(x_I) + b) \tag{5}$$

where $\phi(x_I)$ denotes the deep features extracted by the complete enhanced backbone, and W and b are the weights and biases of the final classification layer, respectively. The choice of EfficientNetV2-L is further motivated by its compound scaling methodology, which efficiently balances network depth, width, and resolution, its high training efficiency, and the strong generalization capabilities afforded by its pretraining on the large-scale ImageNet-21K dataset. These attributes are particularly beneficial for capturing multi-scale species features.

4.2.2. Strategies for addressing long-tailed data distribution

To effectively address the prevalent long-tailed distribution of IAS data, a combined strategy involving data-level adjustments and loss function modification is employed.

Class-aware data resampling. At the data level, to counteract the imbalance in sample numbers across different IAS categories, a class-aware data resampling strategy is implemented during the training phase. This strategy aims to provide the model with a more balanced view of the classes, particularly increasing the exposure to minority (tail) classes. The sampling weight w_c for each class c is determined based on the inverse of its sample frequency N_c:

$$w_c = \frac{1}{N_c}, \quad (c = 1, 2, \ldots, C_{total}) \tag{6}$$

where C_{total} is the total number of classes. These weights are then normalized to obtain the sampling probability p_c^{sample} for each class:

$$p_c^{sample} = \frac{w_c}{\sum_{i=1}^{C_{total}} w_i} \tag{7}$$

This class-aware sampling approach adjusts the data distribution presented to the model during training. Increasing the sampling frequency of minority classes (i.e., oversampling) ensures the model is exposed to a more balanced representation of all species. This process counteracts the model's tendency to overfit to majority classes and significantly improves its ability to learn robust features for rare species.

Gradient-aware focal loss ($\mathcal{L}_{focal}$). Complementing the data resampling strategy, the training of the visual backbone M_E employs an improved Focal Loss function ($\mathcal{L}_{focal}$) at the loss function level. This addresses the gradient imbalance caused by disparities in class sample densities. The Focal Loss is defined as:

$$\mathcal{L}_{focal}(p_t) = -\alpha_c (1 - p_t)^{\gamma} \log(p_t) \tag{8}$$

where p_t is the model's predicted probability for the true class y (i.e., $p_t = p_y$). The class balancing factor α_c is set as the inverse of the number

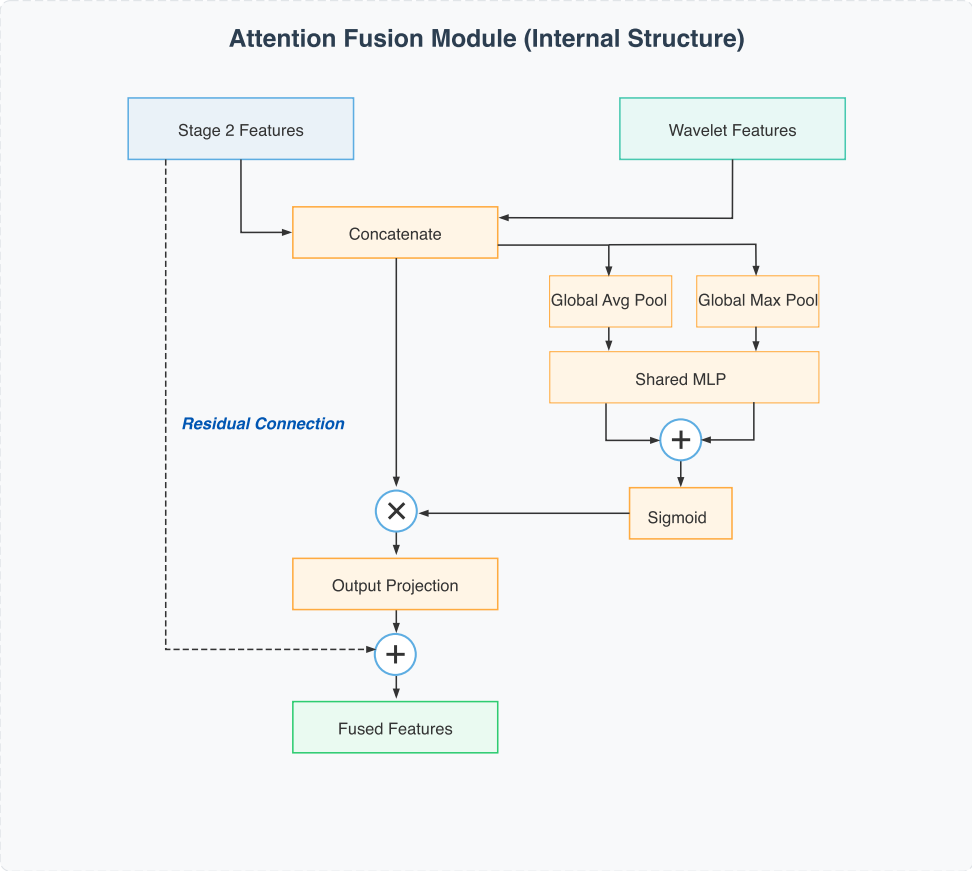


Fig. 7. The internal structure of the Attention Fusion Module. It employs a channel attention mechanism inspired by CBAM and SE-Net, with parallel pooling and a residual connection, to effectively merge backbone features with wavelet-derived features.

of samples for class c (i.e., $\alpha_c = 1/N_c$), providing class-level balance. The focusing parameter γ (set to 2 in this study) modulates the loss, down-weighting easily classified samples (those with high p_t) and thereby focusing training on hard-to-classify, often minority-class examples. The synergistic application of class-aware resampling and Focal Loss provides a robust dual-path optimization to mitigate the adverse effects of long-tailed data distributions.

4.2.3. *Wavelet-enhanced feature extraction for fine-grained recognition*

To enhance the model's ability to distinguish morphologically similar species, an architectural enhancement involving wavelet decomposition and attention-based fusion is incorporated into the visual backbone. This process, illustrated in Fig. 6, enriches the backbone's standard feature maps with multi-scale frequency information.

Wavelet decomposition and attention fusion. As shown on the right path of Fig. 6, the input image first undergoes a 2D Discrete Wavelet Transform (DWT) using a "haar" wavelet. This decomposes the image into four sub-bands representing different frequency components: low-frequency approximations (LL) and high-frequency details in the horizontal (LH), vertical (HL), and diagonal (HH) directions. These sub-bands are concatenated and passed through a shallow convolutional network (Conv) to produce a refined "Wavelet Features" map.

These "Wavelet Features" are then fused with the intermediate features from the main backbone. To effectively merge these heterogeneous feature maps (i.e., spatial features from the CNN and frequency features from the DWT), we designed an Attention Fusion Module. The design of this module is inspired by seminal attention mechanisms, particularly the channel attention branches of Squeeze-and-Excitation Networks (SE-Net) (Hu et al., 2018) and the Convolutional Block Attention Module (CBAM) (Woo et al., 2018). We adapted this structure to specifically address the challenge of fusing our two distinct feature types, leveraging its proven ability to recalibrate channel-wise feature responses. The internal structure of this module is detailed in Fig. 7.

Inside the module, the fusion process unfolds as follows:

1. The "Stage 2 Features" from the backbone and the auxiliary "Wavelet Features" are concatenated along the channel dimension.
2. To generate channel attention weights, the concatenated features are processed along two parallel paths: they are fed into a Global Average Pooling layer and a Global Max Pooling layer, respectively. This allows the module to capture both the overall contextual information (average) and the most salient feature signals (max).
3. The outputs of both pooling layers are then passed through a shared-weight Multi-Layer Perceptron (MLP).
4. The two resulting vectors from the MLP are added together element-wise. A Sigmoid activation function is then applied to the sum to generate the final attention weights, scaled between 0 and 1.
5. These weights are multiplied (broadcasted) channel-wise with the original concatenated feature map, adaptively re-calibrating its channels to emphasize more informative features.
6. The modulated features are passed through an "Output Projection" (a convolutional layer) to refine the fused representation and match the channel dimension of the original backbone features.
7. Finally, a residual connection, a core principle from ResNet (He et al., 2016), adds the original "Stage 2 Features" to the output of the projection layer. This connection is critical for ensuring stable informa-

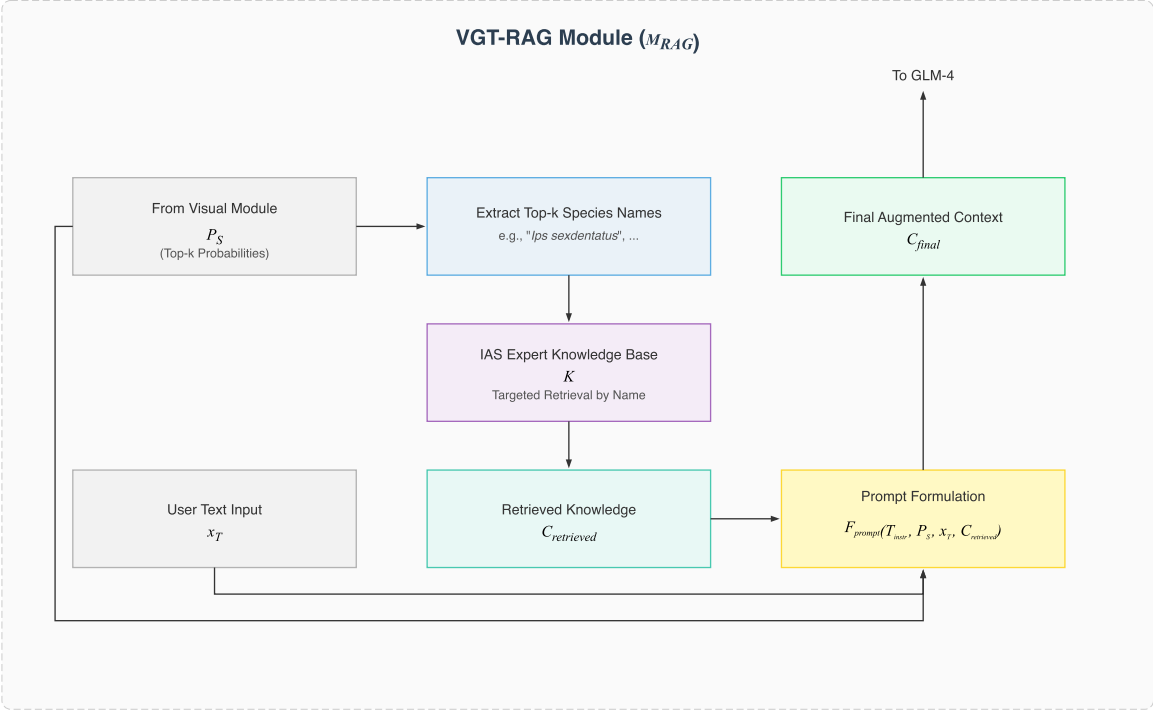


Fig. 8. The architecture of the Vision-Guided Targeted RAG (VGT-RAG) Module (M_{RAG}). The module's key innovation is using the Top-k probabilities (P_S) from the visual model to initiate a targeted retrieval from the IAS Expert Knowledge Base (K). During prompt formulation, the retrieved expert knowledge ($C_{retrieved}$) is synthesized with the visual model's predictions, user text (x_T), and a set of predefined system instructions (T_{instr}). This creates a final, augmented context (C_{final}) tailored for precise reasoning by the final LLM.

tion flow and preventing gradient degradation, allowing the module to function as an additive refinement on top of the original features.

These "Fused Features", now enriched with multi-scale textural and edge information from the wavelet domain, are then passed to the subsequent layers (Stages 3–4) of the EfficientNetV2-L backbone for further processing, ultimately leading to a more robust and fine-grained final prediction.

4.3. vision-guided targeted RAG module (M_{RAG})

Traditional Retrieval Augmented Generation (RAG) approaches, which typically trigger semantic searches of knowledge bases using user-provided text, can face significant challenges in specialized domains such as Invasive Alien Species (IAS) identification. These systems can struggle when user descriptions are imprecise and may retrieve irrelevant information when dealing with morphologically similar species.

To address these limitations, this study introduces the **Vision-Guided Targeted Retrieval Augmented Generation (VGT-RAG)** strategy. This approach leverages the preliminary outputs from the visual recognition module to meticulously guide the knowledge retrieval process, significantly enhancing the relevance and precision of the augmented information. The architecture of the VGT-RAG module is detailed in Fig. 8.

As shown in Fig. 8, the module's operation can be broken down into its core components:

1. **Inputs**: The module receives two primary inputs: the Top-k probability distribution (P_S) from the Visual Recognition Module and the user-provided textual description (x_T).
2. **Targeted knowledge retrieval**: This is the central mechanism of the module. Unlike conventional RAG, the retrieval process is initiated by the visual prior, not user text.
   - First, the Top-k candidate species names are extracted from the probability distribution P_S.
   - These names are then used as direct, precise queries for a **Targeted Retrieval by Name** from the IAS Expert Knowledge Base (K). For example, if "Ips sexdentatus" is a candidate, the system retrieves the specific document "Ips sexdentatus.docx".
   - This process yields a highly relevant block of text, the **Retrieved Knowledge ($C_{retrieved}$)**, which contains expert information only about the visually plausible candidate species.
3. **Prompt formulation**: With the relevant knowledge retrieved, the next step is to assemble all available information into a final context for the LLM. A dedicated function, F_{prompt}, combines the original Top-k probabilities (P_S), the user text (x_T), and the newly retrieved knowledge ($C_{retrieved}$), along with a set of system instructions (T_{instr}), to create the **Final Augmented Context (C_{final})**.

$$C_{final} = F_{prompt}(T_{instr}, P_S, x_T, C_{retrieved}) \tag{9}$$

4. **Output to LLM**: This meticulously constructed context, C_{final}, is then passed to the GLM-4 model for the final inference stage, where it will perform context-aware reasoning to determine the final identification.

By directly using the visual model's hypotheses to focus knowledge retrieval, the VGT-RAG strategy provides highly relevant contextual information to the LLM. This significantly enhances the LLM's ability to perform accurate reasoning, especially in distinguishing between similar species where user descriptions alone might be ambiguous, and reduces the system's sensitivity to the quality of user-provided text.

4.3.1. Prompt engineering for GLM-4

The interaction with the GLM-4 model is guided by a carefully structured prompt, which includes a system-level instructional template (T_{instr}) and user-level content combining visual model outputs, user text, and retrieved knowledge.

The instructional template T_{instr} defines the expert role of the LLM and the strict rules for its output:

```
You are an experienced expert in forestry pest and disease
    control.
Your task is to accurately judge and identify the most
    likely species
name based on the provided background knowledge, the image
    model's
prediction results, and the user's description.
You must strictly follow the rules below for judgment, and
    your answer
must [ONLY CONTAIN THE NAME OF ONE SPECIES], without any
    other text,
explanations, numbering, or polite phrases. For example, if
    the judgment
the result is "Hyphantria cunea larva", your answer should
    be "Hyphantria
cunea larva"; if the judgment result is "Hyphantria cunea",
    your answer
should be "Hyphantria cunea".

Rules:
1. If the image model's identification probability for any
    species
   is >= 90%, directly output the name of that species (
       ignoring
   the species description and background knowledge at this
       point).
   Ensure that the output name is [EXACTLY THE SAME] as the
   highest probability class name; do not make any
       modifications,
   simplifications, or generalizations.
2. If no species' identification probability reaches 90%,
    please make a
   comprehensive judgment based on the following information
       sources:
   a. The image model's Top-3 prediction results and their
      probabilities.
```

```
   b. The user-provided species description (if not empty).
   c. The provided background knowledge (if available).
   Please judge based on this information and output the
       species name
   you deem most appropriate.
3. If the user-provided species description is empty, but
    the background
   knowledge is not empty (and rule 2 applies), then judge
       primarily
   based on the background knowledge and the image model's
       prediction
   results.
4. If both the user-provided species description and
    background
   knowledge are empty (or no relevant knowledge is found) (
       and rule 2
   applies), then directly output the species name with the
       highest
   probability predicted by the image model.
```

The user-level content includes the formatted visual predictions and user text, as generated by the function $F_{format}(P_S, x_T)$:

```
Image model Top-k prediction results:
Class: [ClassName1], Probability: [Prob1]%
Class: [ClassName2], Probability: [Prob2]%
...
User-provided species description:
[UserTextInput]
```

This information, prepended with the aggregated retrieved knowledge ($C_{retrieved}$), forms the complete input for the user role to GLM-4, ensuring it has all necessary data in a clear format to adhere to the predefined reasoning rules.

5. Experiments

To objectively evaluate the effectiveness of the multi-modal fusion identification framework for invasive alien species proposed in this study, we conducted a series of experiments. This chapter first introduces the experimental environment, data processing methods, and evaluation metrics used. Subsequently, it focuses on presenting the performance comparison of baseline models and the ablation study results of core components.

5.1. Data preprocessing

We divided the entire dataset into training and testing sets at an 8:2 ratio. To ensure a balanced distribution of classes within each data subset after partitioning, we employed a stratified sampling strategy, performing an independent 8:2 split within each invasive species category.

Input images were resized to specific resolutions tailored for each model. For the baseline comparison models, we adopted the input resolutions recommended or used in their respective original publications to ensure fair comparison under their optimal settings. For our proposed model, InvaderDefender, images were resized to 480x480 pixels. During the training phase, to enhance the model's generalization ability, we applied a series of data augmentation techniques, including random horizontal flipping, random rotation, and random perturbations to image brightness, contrast, saturation, and hue, to simulate image variations under different lighting and shooting conditions. It is important to note that these data augmentation techniques were applied only during the training phase; during the testing phase, images were only resized (to their model-specific resolution), normalized, and standardized without any augmentation. Subsequently, image pixel values were normalized from the [0, 255] range to the [0, 1] range. Finally, the normalized image data underwent standardization, where the mean was subtracted

from and then divided by the standard deviation for each of the RGB three channels, resulting in pixel values approximately in the [-1, 1] range for network input. The standardization formula is:

$$\text{output} = \frac{\text{input} - \text{mean}}{\text{std}} \tag{10}$$

5.2. Implementation details

In this study, the total number of training epochs for the visual model was set to 50, with a batch size of 16. During the training process, we used the AdamW optimizer for model parameter optimization and enabled mixed-precision training to enhance efficiency. The initial learning rate was set to 1×10^{-4}, and after 20 epochs of training, the learning rate was adjusted to 5×10^{-5}. All experiments recorded the best results as the final performance of the model.

The experimental platform hardware configuration was an Intel Xeon Gold 6430 CPU and an NVIDIA RTX 4090 (24G VRAM) GPU. The software environment consisted of Ubuntu 22.04, Python 3.12, PyTorch 2.5.1, and CUDA 12.4. For the semantic enhancement stage of our InvaderDefender framework, the GLM-4 large language model was deployed and managed via the **Xinference** serving framework (Lu et al., 2024). To provide a comparative baseline in our ablation study (Section 5.5), the basic RAG module was implemented based on the open-source **Langchain-Chatchat** project (Liu et al., 2024).

In terms of runtime performance and computational cost, we benchmarked the system's response time and VRAM usage on our deployment test environment. This environment utilized an NVIDIA RTX 5090 GPU, which was an upgrade from the NVIDIA RTX 4090 used for the initial model training. On this 5090 testbed, the Stage 1 visual recognition module (M_E) achieved an average inference time of approximately 0.186 seconds per image. The Stage 2 semantic enhancement module, which includes RAG retrieval and the GLM-4 LLM inference via the Xinference API, showed an average response time of approximately 1.097 seconds. This results in an average total response time of approximately 1.283 seconds for the complete pipeline. Furthermore, the GLM-4 model introduces significant computational cost in terms of VRAM: it requires 18.7G for deployment and 23G during active inference.

5.3. Evaluation metrics

To comprehensively evaluate the performance of the framework proposed in this study, we adopted the following commonly used classification task evaluation metrics:

- Top-1 Accuracy: The proportion of test samples correctly classified by the model into the category with the highest probability.
- Top-3 Accuracy: The proportion of test samples where the true category is among the three categories with the highest predicted probabilities by the model.
- Precision: For each category, the proportion of samples predicted by the model as belonging to that category that actually belong to that category. This study reports Macro-Precision, which is the arithmetic mean of precision for all categories.
- Recall: For each category, the proportion of samples in the dataset that actually belong to that category and are correctly predicted by the model as belonging to that category. This study reports Macro-Recall, which is the arithmetic mean of recall for all categories.
- F1-Score: The harmonic mean of precision and recall ($2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$), a common metric for comprehensively evaluating model performance. This study reports a Macro-F1 Score.
- Mean Accuracy: The arithmetic mean of the accuracy for each class, which provides a more balanced reflection of the model's overall identification performance across all categories, especially for tail classes with fewer samples.

These metrics will be used to comprehensively compare the performance of the multi-modal fusion identification framework proposed in this study with other models and to verify the effectiveness of the key internal components of the framework through ablation studies.

5.4. Comparison experiments

To comprehensively evaluate the performance of the multi-modal identification framework proposed in this study on the invasive species identification task, we conducted a horizontal comparison with several current mainstream or classic deep learning models on the same dataset. These comparative models include ResNet101 (He et al., 2016), DenseNet121 (Huang et al., 2017), Inception-ResNet-V2 (Szegedy et al., 2017), ViT-Base-Patch16-224 and ViT-Huge-Patch14-224 (Dosovitskiy et al., 2020), the large multimodal model CLIP (Radford et al., 2021), Swin-Base-Patch4-Window7-224 (Liu et al., 2021), EfficientNetV2-L (Tan & Le, 2021) (which serves as the visual backbone basis), SwinV2-Base-Window8-256 (Liu et al., 2022a), ConvNeXt-Base (Liu et al., 2022b), and MobileNetV4-Hybrid-Large (Qin et al., 2024). Detailed performance comparison results are shown in Table 2.

The results presented in Table 2 demonstrate the competitive performance of most modern deep learning architectures. To further situate our work, we conducted an in-depth evaluation using the powerful CLIP (ViT-L/14) model under three distinct settings. The model's **zero-shot performance is merely 21.65 %**, which underscores the substantial domain gap and validates the necessity of domain-specific training for this fine-grained task.

After in-domain training, CLIP proved to be a formidable visual baseline. The **linear probing** approach, which tests the quality of the frozen pre-trained features, achieved a high accuracy of **97.34 %**. Interestingly, **full fine-tuning** resulted in a slightly lower accuracy of **97.04 %**. This highlights a critical insight: for fine-grained tasks, aggressively fine-tuning a large pre-trained model is not always the optimal strategy, as it may lead to overfitting on the specialized dataset and a **consequent loss of generalization ability**, potentially degrading the model's powerful pre-trained features.

Despite the strong performance of these advanced **vision-only baselines**, our proposed **InvaderDefender framework achieved a superior Top-1 accuracy of 99.31 %**. More importantly, it demonstrated marked improvements in metrics that reflect performance on imbalanced data. InvaderDefender's **Macro Recall of 97.11 %** significantly surpasses that of both the linear probe (92.36 %) and fine-tuned (90.43 %) CLIP setups. This comparison critically validates that our two-stage "Visual Recognition-Semantic Enhancement" approach, which leverages both visual data and contextual expert knowledge, is not only more accurate but also more robust than even the most powerful **approaches that rely solely on visual information** for this challenging task.

5.5. Ablation study

To further verify the actual contributions of the various components introduced in this research framework-including the joint processing strategy for data long-tail distribution (class-aware data resampling and gradient-aware focal loss function), the wavelet-enhanced feature extraction module, and the different configurations of the Retrieval Augmented Generation (RAG) module-we designed and conducted a series of ablation experiments. The experimental results are summarized in Table 3. The baseline for these experiments is the EfficientNetV2-L visual model. All RAG configurations in this ablation study utilized the same prompt structure detailed in Section 4.3.1 to ensure a fair comparison of their retrieval and generation augmentation capabilities.

The ablation study primarily includes the following configurations:

- **EfficientNetV2 (Baseline)**: Utilizes only the EfficientNetV2-L visual model for image classification. This model achieved a Mean Accuracy of 94.96 % and an Overall Accuracy (Top-1) of 98.41 %.
- **EfficientNetV2 + Focal loss (FL)**: Augments the baseline model with only the gradient-aware Focal Loss (Section 4.2.2). This config-

Table 2
Performance comparison of different models on invasive species classification. This table presents the evaluation metrics for various deep learning models, ordered by their approximate publication year. Metrics include Top-1 Accuracy, Macro-averaged Precision, Recall, and F1 Score. The highest value in each column is highlighted in bold.

Model Name	Top-1 Accuracy(%)	Macro Precision(%)	Macro Recall(%)	Macro F1 Score(%)
ResNet101	96.91	94.79	92.68	93.55
Inception-ResNet-V2	96.74	96.10	92.81	94.31
DenseNet121	96.79	95.70	93.59	94.59
ViT-Base-Patch16-224	96.91	95.85	92.41	93.90
ViT-Huge-Patch14-224	96.57	94.34	92.33	93.20
CLIP (ViT-L/14, Zero-shot)	21.65	18.75	18.41	12.27
CLIP (ViT-L/14, Linear Probing)	97.34	96.08	92.36	93.92
CLIP (ViT-L/14, Full Fine-tuning)	97.04	95.50	90.43	92.57
Swin-Base-Patch4-Window7-224	97.86	95.72	93.84	94.64
EfficientNetV2-L	98.41	95.19	94.96	94.96
SwinV2-Base-Window8-256	97.64	95.56	92.98	94.14
ConvNeXt-Base	98.24	97.55	94.89	96.05
MobileNetV4-Hybrid-Large	97.60	96.50	94.04	95.08
InvaderDefender(Ours)	**99.31**	**98.88**	**97.11**	**97.94**

uration yielded a Mean Accuracy of 95.49 % and an Overall Accuracy of 98.41 %.

- **EfficientNetV2 + Resampling (RS)**: Augments the baseline model with only class-aware data resampling (Section 4.2.2). This configuration yielded a Mean Accuracy of 95.66 % and an Overall Accuracy of 98.33 %.
- **EfficientNetV2 + Long-tail strategy (FL + RS)**: Augments the baseline model with the joint long-tail distribution processing strategy (both FL and RS). This configuration achieved a Mean Accuracy of 95.76 % and an Overall Accuracy of 98.54 %, demonstrating a synergistic improvement over the individual components.
- **EfficientNetV2 + Wavelet**: Incorporates the wavelet-enhanced feature extraction module (Section 4.2.3) into the baseline model. This resulted in a significant performance boost, achieving a Mean Accuracy of 97.00 % (+2.04 p.p. over baseline) and an Overall Accuracy of 98.67 % (+0.26 p.p. over baseline). This highlights the crucial role of wavelet-derived multi-scale features in improving fine-grained distinction.
- **EfficientNetV2 + RAG (Basic RAG)**: Integrates the baseline visual model with a foundational RAG module. This RAG setup involves constructing a query from the visual model's output probabilities and user text, which is then used for semantic retrieval from the expert knowledge base to augment the context for the LLM, as generally outlined in the initial description of the RAG process (Section 4.3 introduction) but without the specific VGT-RAG enhancements. This configuration achieved a Mean Accuracy of 90.31 % and an Overall Accuracy of 98.28 %. The decrease in Mean Accuracy compared to the baseline suggests that this more general RAG approach might have introduced less relevant or potentially noisy information for some classes, negatively impacting the balanced accuracy.
- **EfficientNetV2 + VGT-RAG**: Employs the proposed VGT-RAG module (detailed in Section 4.3) with the baseline visual model. This configuration achieved a Mean Accuracy of 95.01 % and an Overall Accuracy of 98.71 %. Compared to the baseline visual model, VGT-RAG improved Overall Accuracy (+0.30 p.p.) and slightly improved Mean Accuracy (+0.05 p.p.). More importantly, compared to the "EfficientNetV2 + RAG (Basic RAG)" configuration, VGT-RAG demonstrated a substantial improvement, particularly in Mean Accuracy (+4.70 p.p.), underscoring the benefits of using visual priors to guide knowledge retrieval for more precise contextual augmentation.
- **InvaderDefender(Ours)**: Represents the complete proposed framework, integrating the EfficientNetV2-L visual backbone with the wavelet-enhancement module, the joint long-tail processing strategy (Resampling + Focal Loss), and the VGT-RAG module. This comprehensive model achieved the highest performance with a Mean Accuracy of 97.11 % and an Overall Accuracy of 99.31 %.

Table 3
Ablation study of different modules and strategies on IAS identification performance. Mean accuracy represents the macro-averaged recall. Overall Accuracy is the Top-1 accuracy.

Method	Mean Accuracy (%)	Overall Accuracy (%)
EfficientNetV2 (Baseline)	94.96	98.41
EfficientNetV2 + Focal Loss(FL)	95.49	98.41
EfficientNetV2 + Resampling (RS)	95.66	98.33
EfficientNetV2 + Long-tail Strategy(FL + RS)	95.76	98.54
EfficientNetV2 + Wavelet	97.00	98.67
EfficientNetV2 + RAG	90.31	98.28
EfficientNetV2 + VGT-RAG	95.01	98.71
InvaderDefender(Ours)	**97.11**	**99.31**

Analysis of the ablation study results in Table 3 reveals the following:

- The **Long-tail strategy** components demonstrated their effectiveness in handling class imbalance. Compared to the baseline's Mean Accuracy (94.96 %), **Focal Loss alone** improved it to 95.49 % (+0.53 p.p.), and **Class-aware Resampling alone** achieved a stronger 95.66 % (+0.70 p.p.). The **combined strategy (FL + RS)** yielded the best result in this group at 95.76 % (+0.80 p.p.), confirming a synergistic benefit. As expected, these strategies primarily enhanced the Mean Accuracy by improving performance on tail classes, while having a minimal effect on the overall Top-1 accuracy, which is dominated by head classes.
- The **Wavelet enhancement** provides the most substantial single-component boost to the Mean Accuracy of the visual model, increasing it from 94.96 % to 97.00 %, and also raises Overall Accuracy to 98.67 %. This indicates its strong capability in extracting discriminative fine-grained visual features.
- The **VGT-RAG** module, when added to the baseline, improves Overall Accuracy (from 98.41 % to 98.71 %) and offers a significant advantage over a more basic RAG implementation (Mean Accuracy of 95.01 % for VGT-RAG vs. 90.31 % for basic RAG). This confirms the effectiveness of the vision-guided targeted retrieval in providing relevant contextual information for the LLM.
- The **InvaderDefender(Ours)** framework, which combines all proposed components (Wavelet enhancement, Long-tail Strategy, and VGT-RAG), achieves the best results across both metrics. Compared to the strongest visual-only variant "EfficientNetV2 + Wavelet", the full "Ours" model further improves Mean Accuracy from 97.00 % to 97.11 % and Overall Accuracy from 98.67 % to 99.31 %. This demonstrates the synergistic benefits of integrating advanced visual feature extraction with effective long-tail data handling and precise knowledge-augmented semantic enhancement. The final model

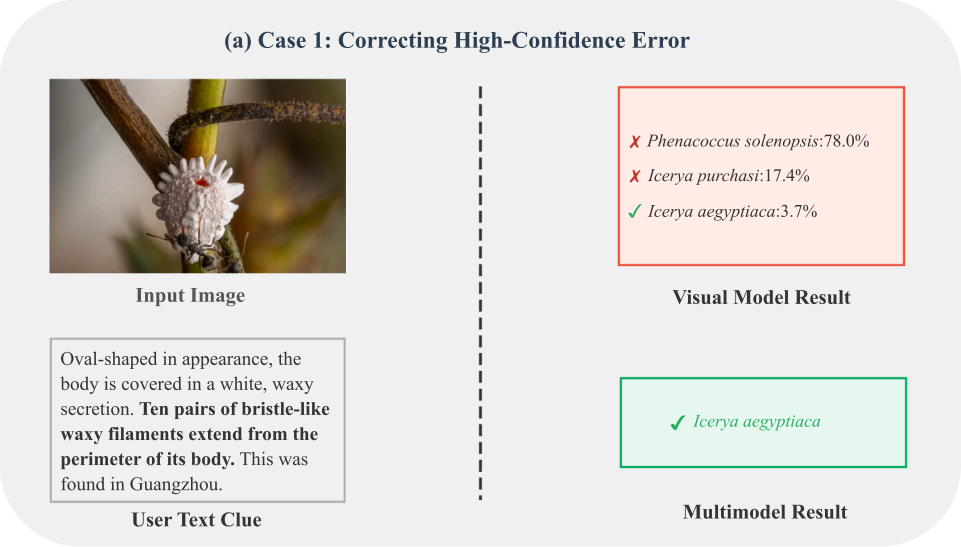


Fig. 9. Case 1: the framework corrects a high-confidence visual error for *Icerya aegyptiaca*. The visual model incorrectly assigned high confidence to *Phenacoccus solenopsis* (78.0 %), while the true species (marked with ✓) received only a 3.7 % probability.

systematically validates the design philosophy of the InvaderDefender framework and the crucial role of each constituent module in achieving superior identification performance.

A detailed per-class accuracy comparison between the baseline model (EfficientNetV2) and the final model (InvaderDefender), which directly demonstrates the performance improvement on rare classes, is provided in Appendix Table A.1.

6. Case study

To qualitatively demonstrate the advanced reasoning capabilities of the InvaderDefender framework, we present three distinct and challenging case studies. These examples, illustrated in Figs. 9–11, were specifically chosen to highlight the system's effectiveness in scenarios where unimodal visual models typically fail: (1) correcting a high-confidence visual misidentification using specific morphological details, (2) overturning a confident visual error based on ecological context, and (3) resolving ambiguity between visually similar species.

Case 1: correcting a high-confidence visual error with morphological reasoning. As shown in Fig. 9, this case presents a formidable challenge where the visual model incorrectly identified the specimen as *Phenacoccus solenopsis* with high confidence (78.0 %). The true species, *Icerya aegyptiaca*, was relegated to a mere 3.7 % probability. However, the user's text provided a decisive clue: a description of "ten pairs of bristle-like waxy filaments" extending from the body's perimeter. This is a textbook diagnostic trait for *I. aegyptiaca*. The VGT-RAG module retrieved expert knowledge confirming this unique feature. The GLM-4 reasoning engine then correctly inferred that this specific morphological evidence was more reliable than the visual model's high-confidence, yet flawed, prediction. It successfully overrode the visual error, elevating the low-probability candidate to the final, correct identification.

Case 2: overturning a visual error with ecological context. This example (Fig. 10) demonstrates the framework's capacity to leverage ecological information to resolve misidentifications. The visual model confidently, yet incorrectly, assigned an 84.2 % probability to *Icerya purchasi*. The user's text, however, focused not on morphology but on the damage caused: the pest "forms galls on tender shoots, petioles, and leaf midribs". Gall formation is a distinct ecological behavior strongly associated with *Leptocybe invasa* but not with the visually predicted species. By integrating this crucial ecological context retrieved from the knowledge base, the InvaderDefender framework concluded that the behavioral evidence was definitive, successfully correcting the high-confidence visual error and identifying the pest as *Leptocybe invasa*.

Case 3: resolving fine-grained ambiguity between similar species. This case (Fig. 11) illustrates a common scenario where the visual model struggles with ambiguity. The model assigned comparable probabilities to two visually similar species: *Corythucha ciliata* (54.4 %) and the correct species, *Icerya purchasi* (42.5 %). The user's text provided the critical details for differentiation, noting the presence of "an ovisac at its rear, and a ridged back"-both classic features of an adult female *I. purchasi*. Armed with this fine-grained information, the GLM-4 was able to break the deadlock. It correctly reasoned that these specific characteristics conclusively matched the description of *I. purchasi* in the expert knowledge base, thereby resolving the visual model's uncertainty and confirming the second candidate as the correct identification.

7. Discussion

The experimental results presented in this study underscore the efficacy of the proposed InvaderDefender framework for the identification of Invasive Alien Species (IAS). Our approach, which integrates advanced visual recognition with semantic enhancement via a Vision-Guided Targeted Retrieval Augmented Generation (VGT-RAG) system, demonstrates superior performance compared to several state-of-the-art unimodal visual models and highlights the benefits of each introduced component.

The Visual Recognition Module (M_E), anchored by the EfficientNetV2-L backbone, provided a strong foundation, as evidenced by its competitive baseline performance (Top-1 Accuracy 98.41 %, Table 2). The incorporation of the joint long-tail strategy, combining class-aware data resampling and gradient-aware Focal Loss, yielded a notable improvement in Mean Accuracy by 0.41 percentage points (from 94.96 % to 95.37 %, Table 3). This confirms the strategy's effectiveness in mitigating the adverse effects of class imbalance inherent in real-world IAS datasets, leading to more equitable performance across common and rare species. Furthermore, the wavelet-enhanced feature extraction module proved to be a critical innovation for the visual component. By decomposing images into multi-resolution sub-bands and fusing these with backbone features via an attention mechanism, the model

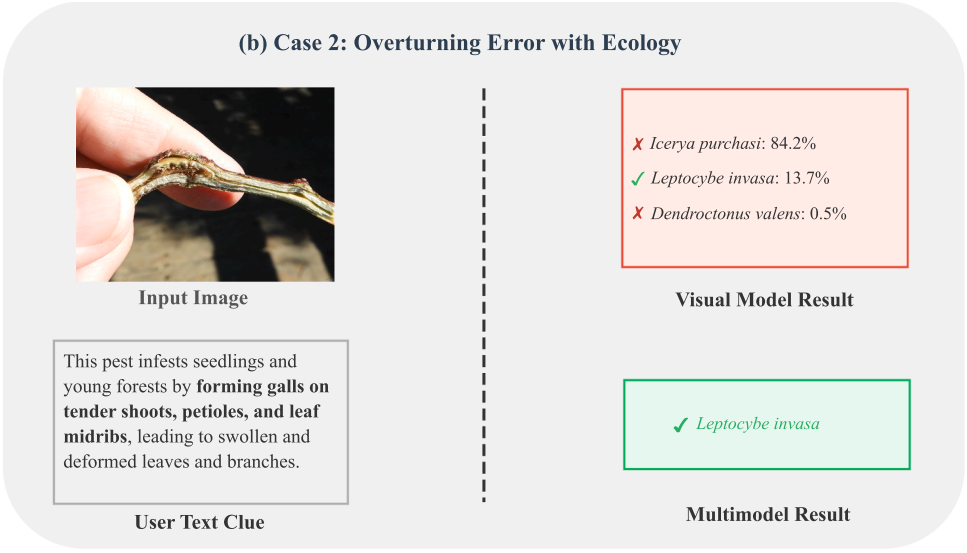


Fig. 10. Case 2: the framework overturns a confident visual misidentification of *Leptocybe invasa* by reasoning about the ecological behavior ("forms galls") described in the text.

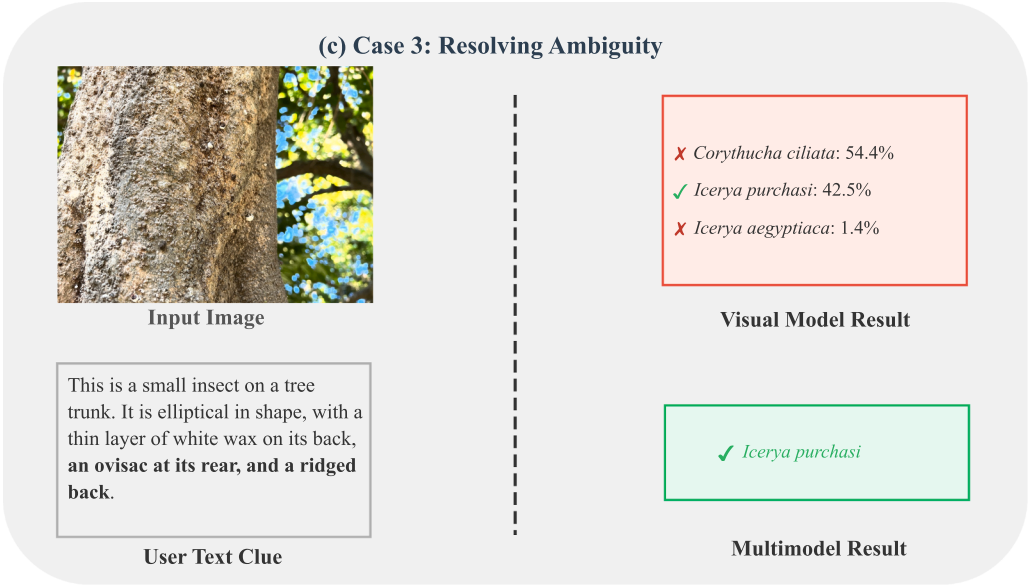


Fig. 11. Case 3: the framework resolves the visual model's ambiguity for *Icerya purchasi* by using the key diagnostic feature of an "ovisac" mentioned in the user's text.

gained an enhanced ability to capture subtle textural and edge details crucial for fine-grained distinctions. This is reflected in the substantial increase in Mean Accuracy by 2.04 percentage points (from 94.96 % to 97.00 %) and Overall Accuracy by 0.26 percentage points (from 98.41 % to 98.67 %) when wavelet enhancement was added to the baseline EfficientNetV2 model (Table 3). This suggests that the richer, multi-faceted feature representations are particularly beneficial for differentiating morphologically similar IAS.

The Semantic Enhancement Module (M_{RAG}) played a pivotal role in refining the initial visual predictions. A key finding is the marked superiority of our proposed VGT-RAG mechanism over a more conventional RAG approach. As shown in Table 3, adding a basic RAG to EfficientNetV2 actually decreased Mean Accuracy from 94.96 % to 90.31 %, likely due to the retrieval of overly broad or irrelevant information from the knowledge base when guided solely by potentially ambiguous user text. In contrast, the VGT-RAG, which uses the Top-k visual predictions to directly and precisely query the IAS expert knowledge base, improved Mean Accuracy to 95.01 % and Overall Accuracy to 98.71 % when added to the baseline. This represents a 4.70 percentage point increase in Mean Accuracy over the basic RAG, underscoring the importance of vision-guided targeted retrieval. This targeted approach ensures that the LLM is augmented with highly relevant, species-specific knowledge, minimizing noise and enhancing its reasoning capabilities, especially in complex cases. The GLM-4, guided by meticulous prompt engineering (Section 4.3.1), effectively acts as an "expert brain", integrating the visual model's confidence scores, user-provided context, and retrieved expert knowledge. The predefined rules, such as deferring to high-confidence (>=90 %) visual predictions, streamline the process while allowing for nuanced judgment in more ambiguous scenarios. The constructed IAS expert knowledge base (Section 3.3) provides the essential domain-specific information that empowers the LLM to perform this informed decision-making.

This contrast highlights a core advantage of our VGT-RAG design and its value in practical applications. The dependency of traditional

RAG on high-quality user text is a significant bottleneck in specialized domains. Our framework fundamentally addresses this challenge by innovatively using the visual model's Top-k predictions as the primary driver for knowledge retrieval, rather than relying solely on user text.

This 'vision-first' strategy establishes a robust, tiered reasoning process. The system's behavior is not brittle; instead, it adapts to the availability and quality of user input. When user text is missing or ambiguous, the framework remains fully operational. Guided by a pre-configured system prompt (e.g., its operational role as 'a forestry expert in China'), the LLM can cross-reference the retrieved expert knowledge for logical consistency. For instance, if a top visual candidate's known distribution is confined to North America, the LLM can detect this geographical contradiction and favor a lower-probability candidate whose distribution includes China. In the absence of such strong contextual evidence, the system gracefully falls back to the visual model's initial probability ranking, ensuring a reliable baseline performance.

When effective user text is provided, it serves as decisive, fine-grained evidence. As demonstrated in our case studies, specific details from the user (e.g., "forms galls on tender shoots") allow the LLM to resolve visual ambiguities with high confidence. Furthermore, the robustness of this tiered approach was validated by the experimental design itself: the `user_input` data was systematically constructed by trained annotators following expert-defined guidelines. This process ensured a diverse range of challenging, real-world descriptions, including texts with varying levels of detail, non-technical language, and potential ambiguities, thus rigorously testing the framework's performance under realistic conditions. Therefore, the VGT-RAG design not only improves identification accuracy but, more importantly, enhances the framework's robustness and feasibility for real-world deployment.

The synergy of the complete InvaderDefender framework is evident when comparing its performance to its constituent parts. The full model, incorporating all enhancements, achieved an Overall Accuracy of 99.31 % and a Mean Accuracy of 97.11 % (Table 3). This is a significant improvement over even the most potent visual-only variant ("EfficientNetV2 + Wavelet", Overall Acc 98.67 %, Mean Acc 97.00 %). Specifically, the full multimodal framework improved Overall Accuracy by 0.64 percentage points and Mean Accuracy by 0.11 percentage points over this strong visual baseline. This demonstrates that the semantic enhancement provided by VGT-RAG offers tangible benefits even when the visual model is already highly optimized, effectively emulating the human expert's process of combining visual cues with domain knowledge. The overall framework's Top-1 Accuracy of 99.31 % (Table 2) also significantly surpasses other tested unimodal deep learning models, further validating our multimodal design philosophy.

The contributions of this research extend beyond performance metrics. The proposal of the "Visual Recognition-Semantic Enhancement" framework offers a novel paradigm for IAS identification. The development of the first IAS multimodal benchmark dataset, including images, structured textual annotations, and a dedicated expert knowledge base for 20 IAS, represents a valuable resource for the research community. This work also provides practical validation for the application of RAG models in specialized biological identification, showcasing how LLMs can be effectively integrated with domain-specific knowledge. The innovative use of Top-k visual outputs not merely as an evaluation metric but as a direct input to guide the retrieval process imbues this information with practical decision-making significance.

The primary objective of this paper is to validate the InvaderDefender framework and demonstrate its performance under controlled, labeled conditions. This validation-focused scope implies inherent limitations. The first is the framework's **closed-set behavior**: trained on a defined set of 20 species, it will incorrectly force any non-target organism into one of the known classes. The second is an **error-propagation risk** due to its serial dependency: if the visual model omits the true species from its initial Top-k, the downstream VGT-RAG stage can only refine those candidates rather than generate new hypotheses.

Addressing the **closed-set limitation** for practical deployment necessitates scaling the dataset. However, our ongoing work to scale the dataset to approximately 100 species (including representative native and non-target taxa) has already exposed two significant practical bottlenecks: (i) an intensified **long-tailed distribution** that threatens Top-k stability for rare classes, and (ii) the **manual labor bottleneck** inherent in the knowledge base construction process has proven substantial.

To address this comprehensive set of challenges-both the inherent limitations and the scalability bottlenecks-our future work will pursue an integrated strategy. First, building upon the expanded dataset, we will implement advanced **open-set recognition (OSR)** capabilities to avoid the forced classification of unknown species. Second, we will investigate **hierarchical classification architectures** specifically designed to mitigate the **error propagation risk** and improve recall for tail classes. Finally, to tackle the scalability bottleneck of knowledge curation, we will leverage **Large Language Models (LLMs)** for semi-automated knowledge base construction, with human-in-the-loop verification to preserve quality.

Furthermore, while the Xinference API facilitates the use of powerful LLMs like GLM-4, the computational demands of such models might be a consideration for resource-constrained deployment scenarios. Our performance benchmarks (reported in Section 5.2) confirm this challenge. The semantic module's high computational cost (23 GB VRAM and 1.1s inference latency) makes direct mobile deployment unfeasible. This strongly supports a hybrid model where the efficient visual module (Stage 1) runs on-device, and the heavy semantic module (Stage 2) is offloaded to a server. The interpretability of LLM decisions, although partially addressed by structured prompts and rules, remains an area for further exploration. Future work could also investigate end-to-end multimodal models that learn to fuse visual and textual information more directly, potentially offering further performance gains or efficiencies. Exploring the integration of other data modalities, such as acoustic data for insect identification or environmental parameters, could also lead to more robust and context-aware IAS identification systems. Finally, real-world field testing and iterative refinement based on user feedback will be crucial for translating this research into a practical tool for IAS management. The present paper, therefore, reports a rigorous performance validation of the proposed architecture while documenting a clear and intended path toward practical, real-world deployment.

8. Conclusion

This paper introduced InvaderDefender, a novel two-stage multimodal fusion framework designed to address the complex challenges of Invasive Alien Species (IAS) identification. By synergistically combining an advanced visual recognition module with a Vision-Guided Targeted Retrieval Augmented Generation (VGT-RAG) based semantic enhancement module, InvaderDefender aims to emulate the comprehensive decision-making process of human experts. The visual module leverages an EfficientNetV2-L backbone, enhanced with a wavelet decomposition module for fine-grained feature extraction and a joint strategy of class-aware resampling and Focal Loss to tackle long-tailed data distributions. The semantic module utilizes the GLM-4 large language model, augmented by an IAS expert knowledge base that is queried in a targeted manner based on the visual model's initial predictions.

Our extensive experiments, conducted on a newly constructed multimodal benchmark dataset for 20 IAS, demonstrate the superiority of the InvaderDefender framework. It achieved a state-of-the-art Top-1 Accuracy of 99.31 % and a Mean Accuracy of 97.11 %, significantly outperforming various mainstream unimodal deep learning models. Ablation

studies meticulously validated the contribution of each component: the wavelet enhancement substantially improved visual feature representation for fine-grained tasks (increasing Mean Accuracy by 2.04 p.p. over baseline visual model), the long-tail strategy effectively balanced performance across classes (increasing Mean Accuracy by 0.41 p.p.), and the VGT-RAG module proved crucial for precise knowledge augmentation, markedly outperforming a basic RAG approach and contributing to the final model's superior accuracy (improving Overall Accuracy by 0.64 p.p. over the best visual-only configuration). These findings confirm that integrating visual perception with knowledge-driven semantic reasoning leads to more accurate and robust IAS identification.

The significance of this research lies not only in the performance achieved but also in its broader implications. InvaderDefender offers a promising and technically robust solution for the timely and accurate identification of IAS, a critical step in their effective management and the mitigation of their ecological and economic impacts. The construction of the multimodal IAS dataset and expert knowledge base provides valuable resources for future research in this domain. Furthermore, this work presents a new paradigm for applying multimodal learning and RAG-empowered LLMs to complex biological identification tasks, highlighting their potential in biodiversity monitoring and other specialized scientific fields.

Future research will focus on expanding the dataset and knowledge base to cover a wider array of species and geographical contexts, exploring end-to-end multimodal architectures, and field-testing the InvaderDefender system for practical deployment. We believe this study lays a strong foundation for the continued development of intelligent systems that can significantly aid in the global challenge of managing invasive alien species.

CRediT authorship contribution statement

Wenda Luo: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Validation, Visualization, Writing – original draft, Writing – review & editing; **Xiaoyu Zhang:** Supervision, Project administration, Writing – review & editing; **Zhibo Chen:** Funding acquisition, Project administration, Resources, Supervision; **Guangyu Huo:** Resources, Supervision, Writing – review & editing; **Rui Zhang:** Data curation, Investigation, Data curation, Investigation; **Liping Mu:** Supervision.

Data availability

The data and code that support the findings of this study are available from the corresponding author, Dr. Xiaoyu Zhang (xyzhang0606@bjfu.edu.cn), upon reasonable request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This research was funded by the National Key Research and Development Program of China (2023YFC2605200).

Appendix A. Full class-wise performance comparison

Table A.1
Class-wise accuracy (Recall) comparison between the baseline model (EfficientNetV2) and the final model (InvaderDefender) for all 20 species. Improvement (p.p.) refers to percentage points.

Species Name	Total Images	Baseline Accuracy (%)	Ours Accuracy (%)	Improvement (p.p.)
Rhabdoscelus lineaticollis	53	90.91	81.82	-9.09
Brontispa longissima	64	92.31	92.31	0.00
Sinoxylon japonicum	68	85.71	92.86	+7.15
Acanthoscelides macrophthalmus	77	81.25	87.50	+6.25
Aulacaspis yasumatsui	87	100.00	100.00	0.00
Phenacoccus solenopsis	101	95.24	100.00	+4.76
Ips sexdentatus	106	77.27	95.45	+18.18
Heterobostrychus aequalis	172	100.00	100.00	0.00
Hylurgus ligniperda	220	90.70	95.35	+4.65
Leptocybe invasa	253	96.08	100.00	+3.92
Hyphantria cunea	516	98.06	100.00	+1.94
Dendroctonus valens	636	100.00	100.00	0.00
Sirex noctilio	774	98.71	100.00	+1.29
Icerya aegyptiaca	855	98.25	99.42	+1.17
Rhynchophorus ferrugineus	923	98.92	100.00	+1.08
Icerya purchasi	946	97.88	98.94	+1.06
Hyphantria cunea larva	1002	98.51	99.01	+0.50
Lantana camara	1028	100.00	100.00	0.00
Corythucha ciliata	1064	99.53	100.00	+0.47
Obolodiplosis robiniae	2688	99.81	99.63	-0.18
Overall Mean Accuracy (20 Classes)	-	94.96	97.11	+2.15

References

Cabezas, M., Kentsch, S., Tomhave, L., Gross, J., Caceres, M. L. L., & Diez, Y. (2020). Detection of invasive species in wetlands: Practical DL with heavily imbalanced data. *Remote Sensing*, *12*(20), 3431.

Cardoso, A. S., Malta-Pinto, E., Tabik, S., August, T., Roy, H. E., Correia, R., Vicente, J. R., & Vaz, A. S. (2024). Can citizen science and social media images support the detection of new invasion sites? a deep learning test case with cortaderia selloana. *Ecological Informatics*, *81*, 102602.

Chen, Y., Qiao, X., Qin, F., Huang, H., Liu, B., Li, Z., Liu, C., Wang, Q., Wan, F., Qian, W., & Huang, Y. (2024). Ipmcnet: A lightweight algorithm for invasive plant multiclassification. *Agronomy*, *14*(2), 333.

Crisigiovanni, E. L., Filho, A. F., Pesck, V. A., & de Lima, V. A. (2021). Potential of machine learning and worldview-2 images for recognizing endangered and invasive species in the atlantic rainforest. *Annals of Forest Science*, *78*(2), 54.

Demertzis, K., & Iliadis, L. (2017). Detecting invasive species with a bio-inspired semi-supervised neurocomputing approach: The case of lagocephalus sceleratus. *Neural Computing and Applications*, *28*, 1225–1234.

Dosovitskiy, A., Beyer, L., Kolesnikov, A., Weissenborn, D., Zhai, X., Unterthiner, T., Dehghani, M., Minderer, M., Heigold, G., Gelly, S. et al. (2020). An image is worth 16x16 words: Transformers for image recognition at scale. *arXiv preprint arXiv:2010.11929*.

Gao, Y., Xue, X., Qin, G., Li, K., Liu, J., Zhang, Y., & Li, X. (2024). Application of machine learning in automatic image identification of insects-a review. *Ecological Informatics*, (p. 102539).

GLM, T., Zeng, A., Xu, B., Wang, B., Zhang, C., Yin, D., Zhang, D., Rojas, D., Feng, G., Zhao, H. et al. (2024). Chatglm: A family of large language models from glm-130b to glm-4 all tools. *arXiv preprint arXiv:2406.12793*.

Gupta, S., Ranjan, R., & Singh, S. N. (2024). A comprehensive survey of retrieval-augmented generation (rag): Evolution, current landscape and future directions. *arXiv preprint arXiv:2410.12837*.

He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep residual learning for image recognition. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 770–778).

Hebert, P. D. N., Cywinska, A., Ball, S. L., & DeWaard, J. R. (2003). Biological identifications through DNA barcodes. *Proceedings of the Royal Society of London. Series B: Biological Sciences*, *270*(1512), 313–321.

Hu, J., Shen, L., & Sun, G. (2018). Squeeze-and-excitation networks. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 7132–7141).

Huang, G., Liu, Z., Van Der Maaten, L., & Weinberger, K. Q. (2017). Densely connected convolutional networks. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 4700–4708).

Hünicken, L. A., Paolucci, E. M., Lavinia, P. D., & Sylvester, F. (2024). Morphological and genetic assessment of invasive corbicula lineages in Southern South America: A case study in argentina. *Animals*, *14*(13), 1843.

Jensen, T., Seerup Hass, F., Seam Akbar, M., Holm Petersen, P., & Jokar Arsanjani, J. (2020). Employing machine learning for detection of invasive species using sentinel-2 and aviris data: The case of kudzu in the united states. *Sustainability*, *12*(9), 3544.

Jia, Y., Ji, X., Wang, X., Zhang, H., Meng, Z., Zhang, J., Chen, Z., Zhuang, P., Xu, D., Guo, W., Yan, Y., & Wei, J. (2025). Qibo: A large language model for traditional chinese medicine. *Expert Systems with Applications*, *284*, 127672. https://doi.org/10.1016/j.eswa.2025.127672

Kotsiantis, S. B., Zaharakis, I., & Pintelas, P. (2007). Supervised machine learning: A review of classification techniques. *Emerging Artificial Intelligence Applications in Computer Engineering*, *160*(1), 3–24.

Lake, T. A., Briscoe Runquist, R. D., & Moeller, D. A. (2022). Deep learning detects invasive plant species across complex landscapes using worldview-2 and planetscope satellite imagery. *Remote Sensing in Ecology and Conservation*, *8*(6), 875–889.

LeCun, Y., Bengio, Y., & Hinton, G. (2015). Deep learning. *Nature*, *521*(7553), 436–444.

Lewis, P., Perez, E., Piktus, A., Petroni, F., Karpukhin, V., Goyal, N., Küttler, H., Lewis, M., Yih, W.-t., Rocktäschel, T., Riedel, S., & Kiela, D. (2020). Retrieval-augmented generation for knowledge-intensive nlp tasks. *Advances in Neural Information Processing Systems*, *33*, 9459–9474.

Lin, T.-Y., Goyal, P., Girshick, R., He, K., & Dollár, P. (2017). Focal loss for dense object detection. In *Proceedings of the IEEE international conference on computer vision* (pp. 2980–2988).

Liu, Q., Song, J., Huang, Z., Zhang, Y., Glide the, & liunux4odoo (2024). langchain-chatchat. https://github.com/chatchat-space/Langchain-Chatchat.

Liu, Z., Hu, H., Lin, Y., Yao, Z., Xie, Z., Wei, Y., Ning, J., Cao, Y., Zhang, Z., Dong, L., Wei, F., & Guo, B. (2022a). Swin transformer v2: Scaling up capacity and resolution. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 12009–12019).

Liu, Z., Lin, Y., Cao, Y., Hu, H., Wei, Y., Zhang, Z., Lin, S., & Guo, B. (2021). Swin transformer: Hierarchical vision transformer using shifted windows. In *Proceedings of the IEEE/CVF international conference on computer vision* (pp. 10012–10022).

Liu, Z., Mao, H., Wu, C.-Y., Feichtenhofer, C., Darrell, T., & Xie, S. (2022b). A convnet for the 2020s. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 11976–11986).

Lu, W., Xiong, L., Zhang, F., Qin, X., & Chen, Y. (2024). Xinference: Making large model serving easy. In *Proceedings of the 2024 conference on empirical methods in natural language processing: System demonstrations* (pp. 291–300). Miami, Florida, USA: Association for Computational Linguistics. https://aclanthology.org/2024.emnlp-demo.30.

Martínez-González, Á. T., Ramírez-Rivera, V. M., Caballero-Vázquez, J. A., & Jáuregui, D. A. G. (2021). Deep learning algorithm as a strategy for detection an invasive species in uncontrolled environment. *Reviews in Fish Biology and Fisheries*, *31*(4), 909–922.

Pyšek, P., Hulme, P. E., Meyerson, L. A., Smith, G. F., Boatwright, J. S., Crouch, N. R., Figueiredo, E., Foxcroft, L. C., Jarošík, V., Richardson, D. M., Suda, J., & Wilson, J. R. U. (2013). Hitting the right target: Taxonomic challenges for, and of, plant invasions. *AoB Plants*, *5*, plt042.

Qin, D., Leichner, C., Delakis, M., Fornoni, M., Luo, S., Yang, F., Wang, W., Banbury, C., Ye, C., Akin, B., Aggarwal, V., Zhu, T., Moro, D., & Howard, A. (2024). Mobilenetv4: Universal models for the mobile ecosystem. In *European conference on computer vision* (pp. 78–96). Springer.

Radford, A., Kim, J. W., Hallacy, C., Ramesh, A., Goh, G., Agarwal, S., Sastry, G., Askell, A., Mishkin, P., Clark, J., Krueger, G., & Sutskever, I. (2021). Learning transferable visual models from natural language supervision. In *International conference on machine learning* (pp. 8748–8763). PmLR.

Roy, H. E., Pauchard, A., Stoett, P., Renard Truong, T., Bacher, S., Galil, B. S., Hulme, P. E., Ikeda, T., Sankaran, K., McGeoch, M. A., Meyerson, L. A., Nuñez, M. A., Ordonez, A., Rahlao, S. J., Schwindt, E., Seebens, H., Sheppard, A. W., & Vandvik, V. (2024). Ipbes invasive alien species assessment: Summary for policymakers. https://doi.org/10.5281/zenodo.11254974

Shoaib, M., Shah, B., Ei-Sappagh, S., Ali, A., Ullah, A., Alenezi, F., Gechev, T., Hussain, T., & Ali, F. (2023). An advanced deep learning models-based plant disease detection: A review of recent research. *Frontiers in Plant Science*, *14*, 1158933.

Sun, C., Shrivastava, A., Singh, S., & Gupta, A. (2017). Revisiting unreasonable effectiveness of data in deep learning era. In *Proceedings of the IEEE international conference on computer vision* (pp. 843–852).

Szegedy, C., Ioffe, S., Vanhoucke, V., & Alemi, A. A. (2017). Inception-v4, inception-resnet and the impact of residual connections on learning. In *Proceedings of the thirty-first AAAI conference on artificial intelligence* AAAI'17 (p. 4278–4284). AAAI Press.

Tan, M., & Le, Q. (2021). Efficientnetv2: Smaller models and faster training. In *International conference on machine learning* (pp. 10096–10106). PMLR.

Wei, X.-S., Song, Y.-Z., Mac Aodha, O., Wu, J., Peng, Y., Tang, J., Yang, J., & Belongie, S. (2021). Fine-grained image analysis with deep learning: A survey. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, *44*(12), 8927–8948.

Woo, S., Park, J., Lee, J.-Y., & Kweon, I. S. (2018). Cbam: Convolutional block attention module. In *Proceedings of the European conference on computer vision (ECCV)* (pp. 3–19).

Xiao, Y., Greiner, R., & Lewis, M. A. (2018). Evaluation of machine learning methods for predicting eradication of aquatic invasive species. *Biological Invasions*, *20*, 2485–2503.

Xie, Y., Gong, Q., Luan, X., Yan, J., & Zhang, J. (2023). A survey of fine-grained visual categorization based on deep learning. *Journal of Systems Engineering and Electronics*, *35*(6), 1337–1356.